TED UNIVERSITY

Department of Computer Engineering

CMPE 491 – Capstone Design Project I | 2026–2027

PerDrobe Project Proposal

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Project name: PerDrobe **Project webpage:** https://perdrobe2026-commits.github.io/

Team Members

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Project Description

PerDrobe is an AI-based personalized wardrobe and clothing recommendation application proposed for our CMPE 491–492 project. It aims to help users recreate styles they discover and assemble outfits from clothes they already own. Users will build a private digital wardrobe by uploading photographs of their clothes. The system will isolate garments through background removal and organize the resulting images for use in recommendations. Explicit style and fit preferences, together with likes, saved outfits and followed accounts, will inform personalization. Users will also be able to combine wardrobe items into outfits. Outfit Remix will adapt discovered outfits by replacing selected pieces with compatible alternatives from the user's wardrobe. Wardrobe-aware shopping will suggest new items that complement owned clothing. Size guidance will use optional body measurements, usual sizes and available product size charts, and will be provided only when sufficient information is available.

The technical focus is a recommendation system that identifies candidate items, assesses their compatibility and ranks them according to each user's preferences. Processing uploaded garment images will support the digital representation of owned clothes and their use in outfit generation. We plan to compare three approaches: a baseline using clothing attributes, a configuration that also uses user preferences and interaction signals, and a configuration that additionally considers the user's wardrobe. Evaluation will cover recommendation relevance, user-rated usefulness, diversity, catalogue coverage and response time. Size and fit guidance will be evaluated only where suitable reference data is available. The application and experiments are planned work; no implementation or experimental results are claimed.